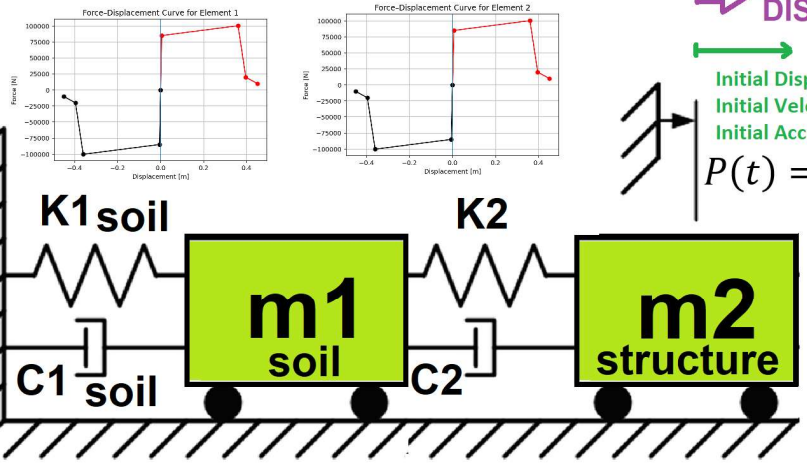
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

# NONLINEAR DYNAMIC ANALYSIS OF A MULTI-DEGREE-OF-FREEDOM (MDOF) SOIL STRUCTURE SYSTEM UTILIZING 50 GROUND MOTIONS IN OPENSEES AND EVALUATION OF SEISMIC RESPONSE SPECTRUM

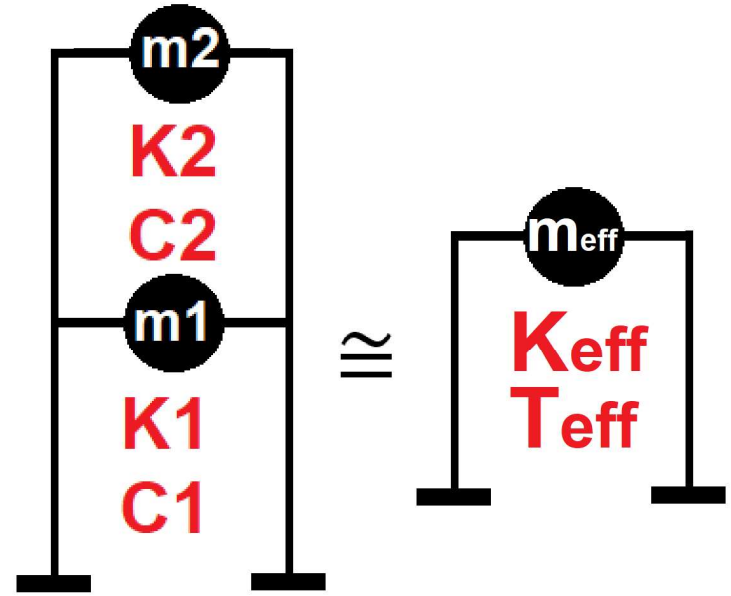
WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)




INCREMENTAL DISPLACEMENT

Initial Displacement
Initial Velocity
Initial Acceleration

$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$







$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

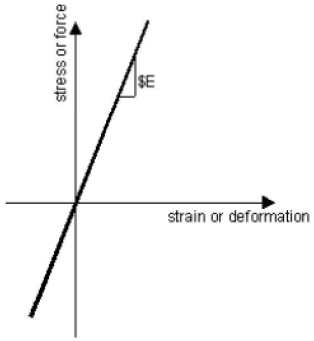
$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$

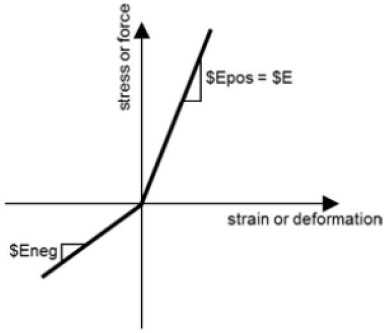
$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$

$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$

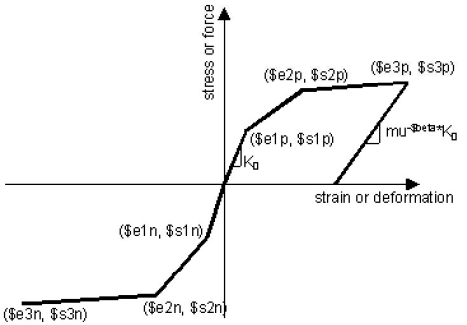
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$




**ELASTIC MATERIAL**




**INELASTIC MATERIAL**

$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...OIL_STRUCTURE_MDOF_RESPONSE_SPECTRUM_EDCI_EVDR.py

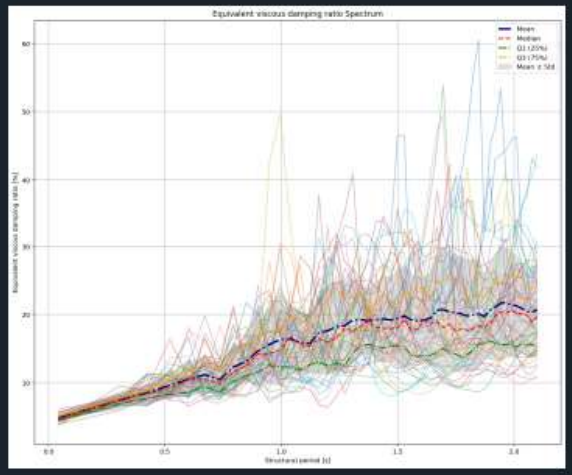
SOIL_STRUCTURE_MDO...CTRUM_EDCI_EVDR.py

```
##########################################################################################################
#                    >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
#     FRAGILITY ANALYSIS BASED ON ACCELERATION, STRUCTURAL DUCTILITY DAMAGE INDEX, ENERGY DISSIPATION
#          CAPACITY INDEX, EQULIVALENT VISCOUS DAMPING RATIO WITH NONLINEAR DYNAMIC ANALYSIS
#    OF A MULTI-DEGREE-OF-FREEDOM (MDOF) SOIL STRUCTURE SYSTEM UTILIZING 50 GROUND MOTIONS IN OPENSEES
#---------------------------------------------------------------------------------------------------------
#                                  INELASTIC RESPONSE SPECTRUM ANALYSIS
#---------------------------------------------------------------------------------------------------------
#               EQUIVALENT VISCOUS DAMPING RATIO: xi_eq = 100 * E_d / (4 * pi * E_s)
#---------------------------------------------------------------------------------------------------------
#          ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacement)
#---------------------------------------------------------------------------------------------------------
# This program performs  Dynamic Analysis on a Multi-Degree-of-Freedom (MDOF) system
# subjected to 50 seismic ground motions. The analysis evaluates the structural response under varying
# levels of seismic intensity.
# The framework is designed to support researchers and engineers in assessing the probabilistic seismic
# performance of structures, with a focus on understanding the impact of uncertainty on structural
# response and design.
#---------------------------------------------------------------------------------------------------------
# Key Features:
# - Simulation of MDOF system using OpenSees.
# - Scaling of ground motions for Nonlinear Dynamic Analysis.
# - Probabilistic fragility assessment based on predefined damage states.
# - Visualization of structural response and fragility curves.
# - Export of results for further analysis.
#---------------------------------------------------------------------------------------------------------
#                      THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
#                                  EMAIL: salar.d.ghashghaei@gmail.com
##########################################################################################################

"""
This code implements a comprehensive nonlinear dynamic analysis framework for
performance-based earthquake engineering assessment of multi-degree-of-freedom
(MDOF) systems. The methodology combines traditional nonlinear time-history
```

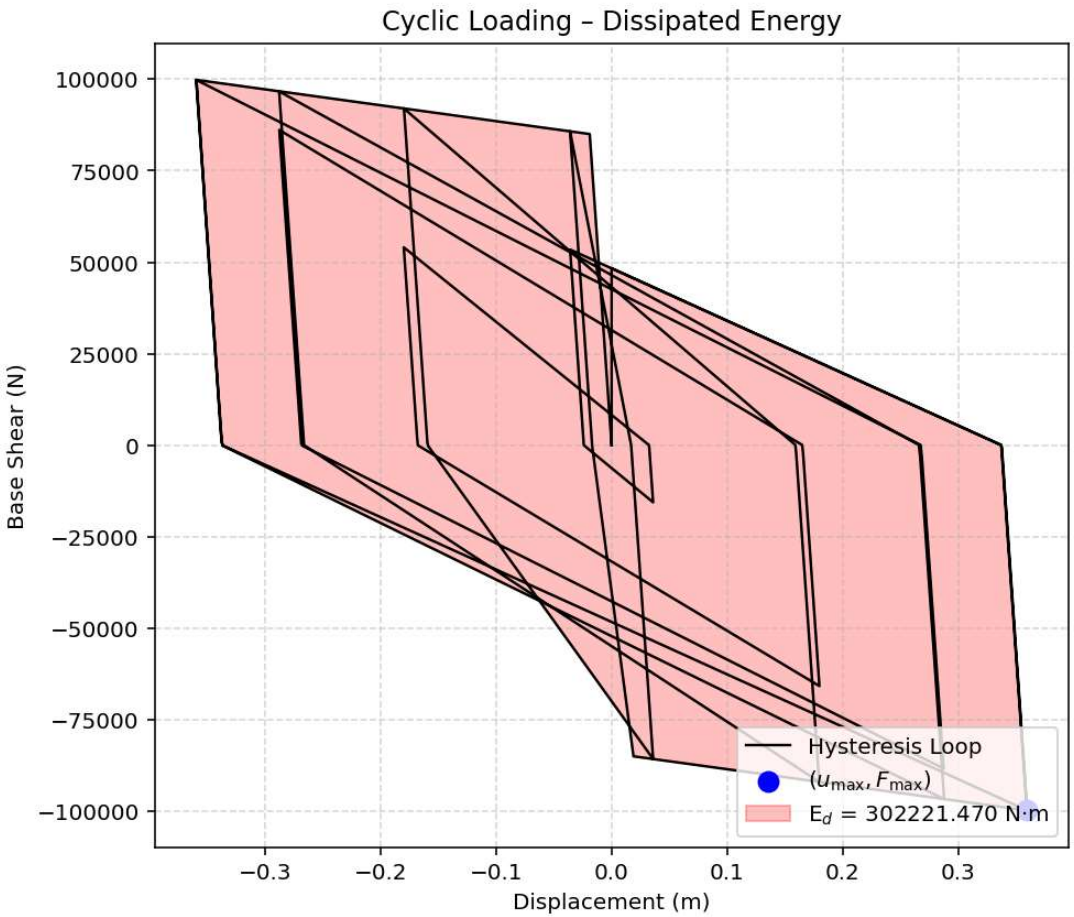
..._SOIL_STRUCTURE_MDOF_8_ANA\+23_SOIL_STRUCTURE_MDOF_8_ANA

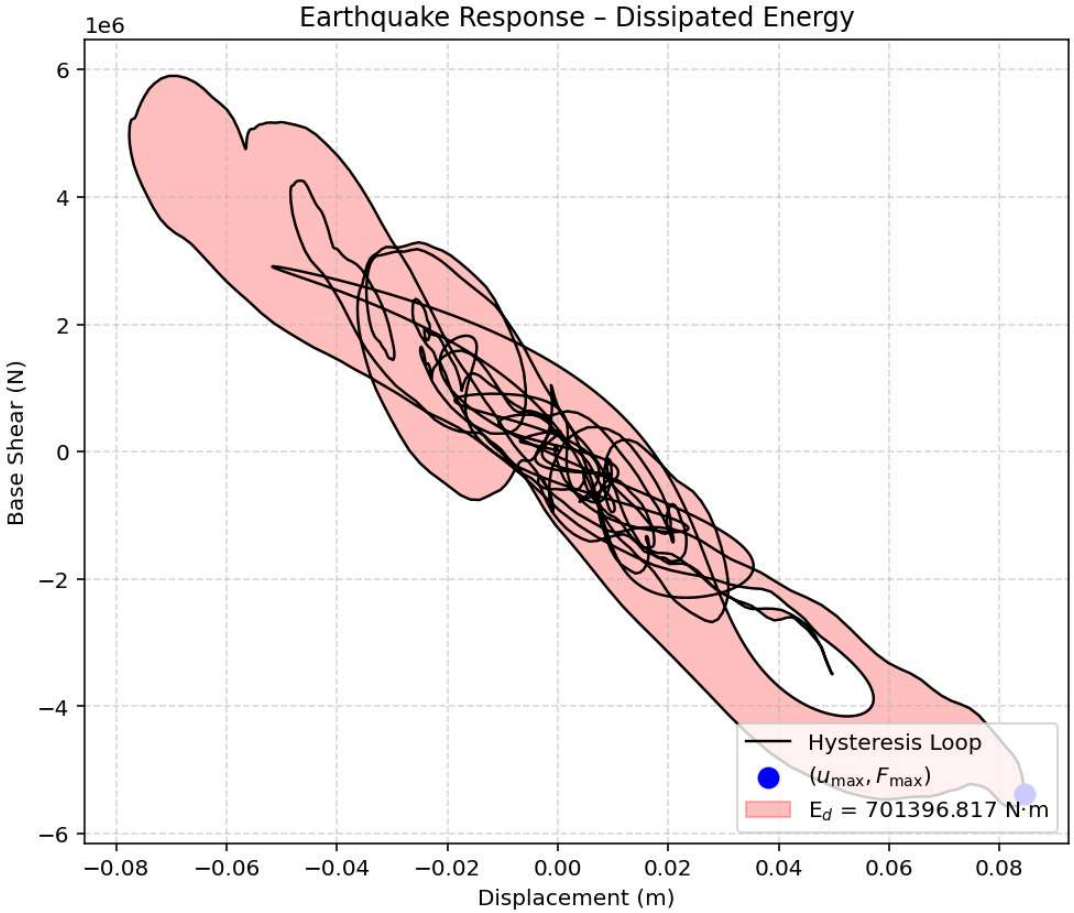
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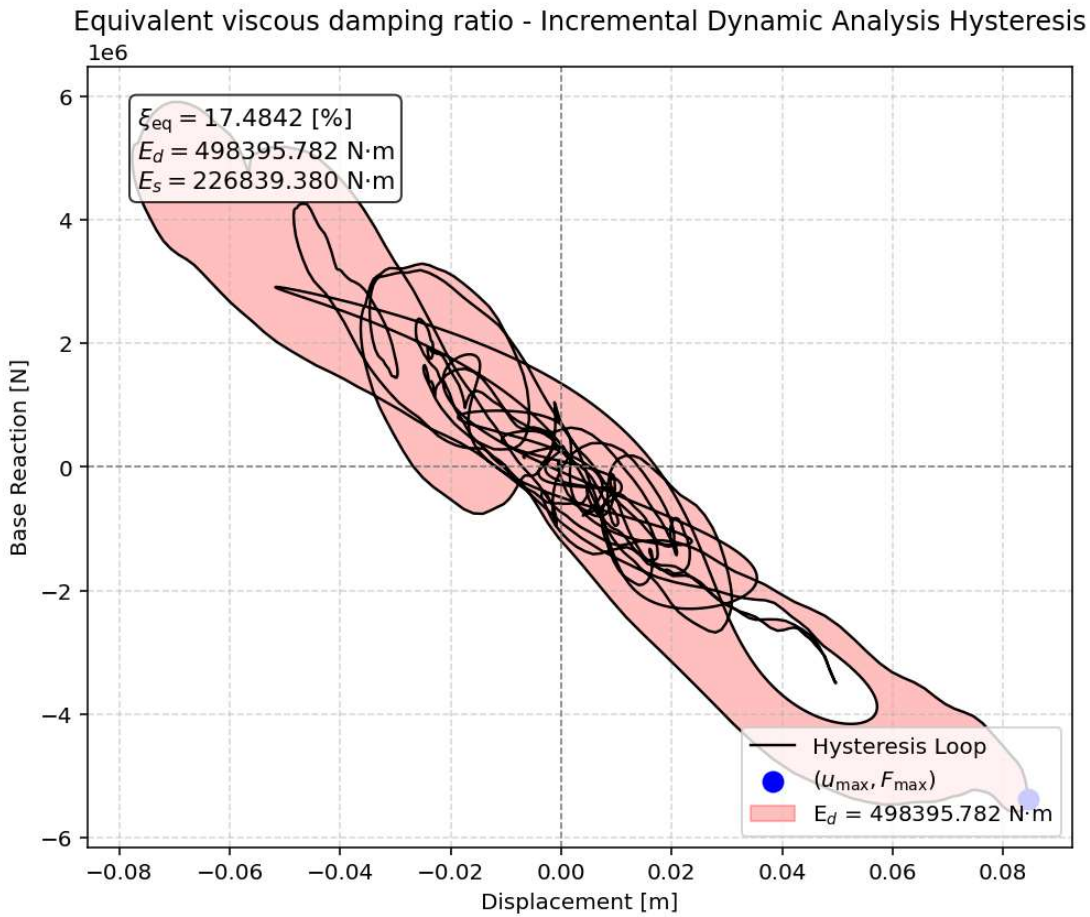


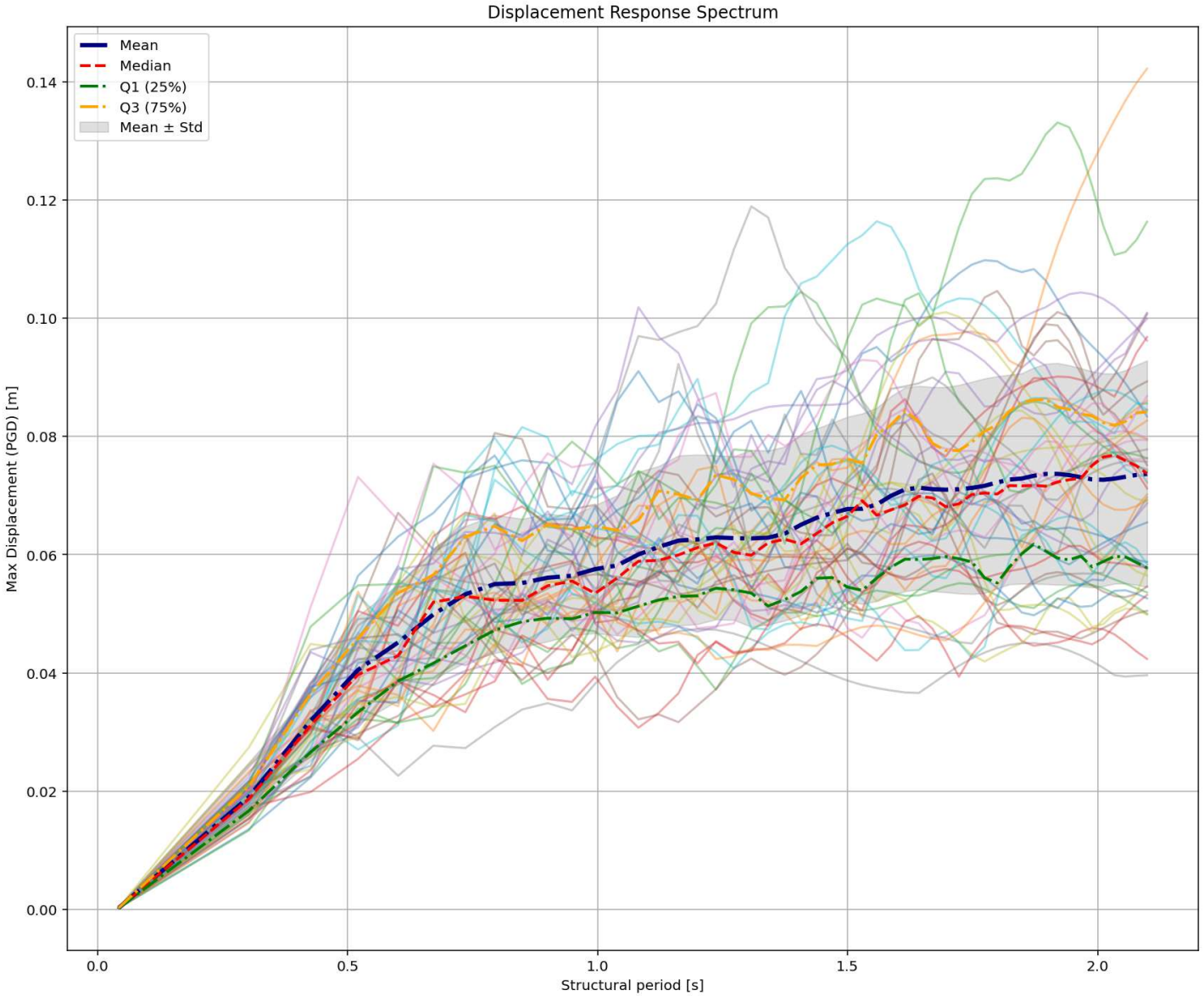
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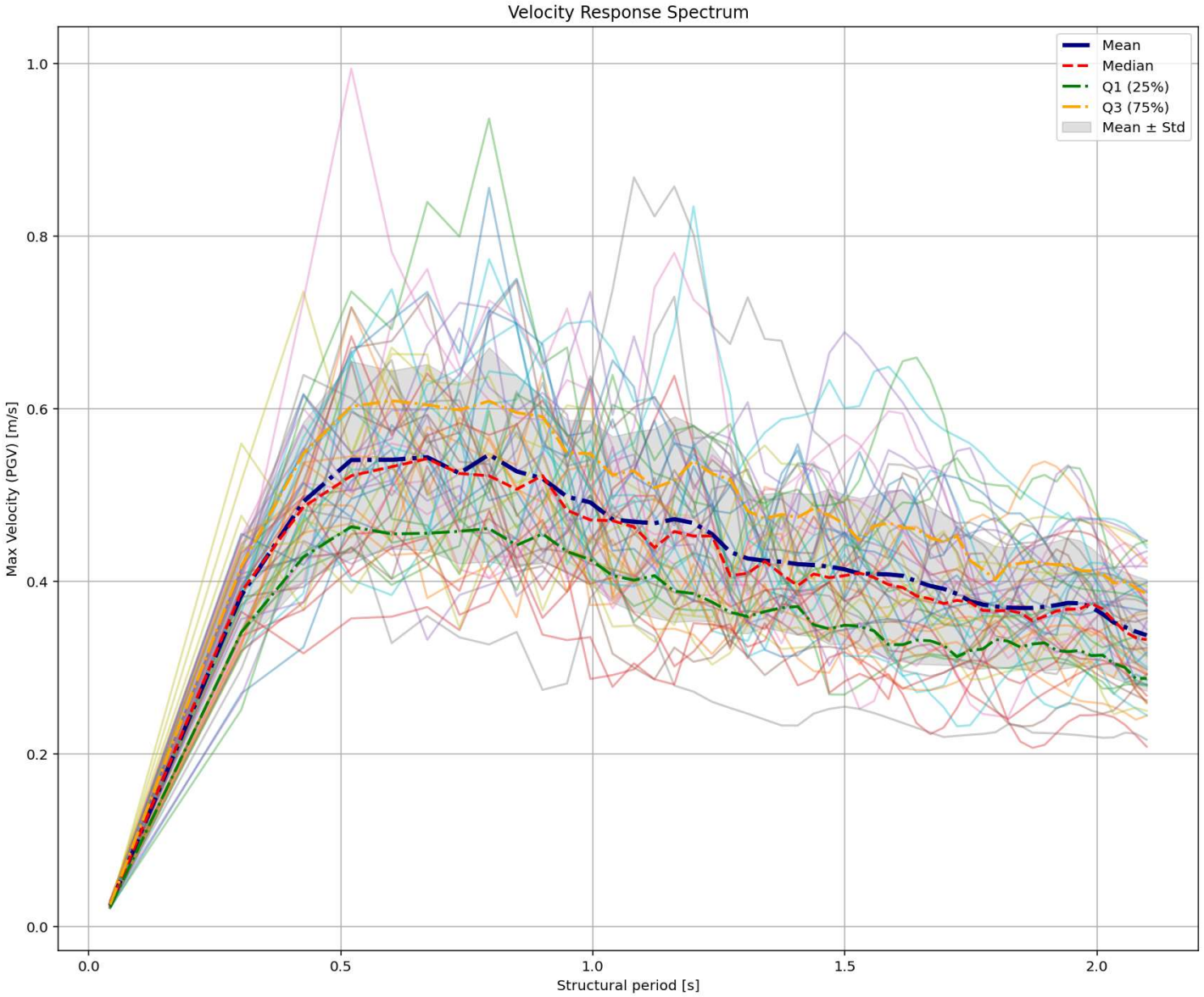
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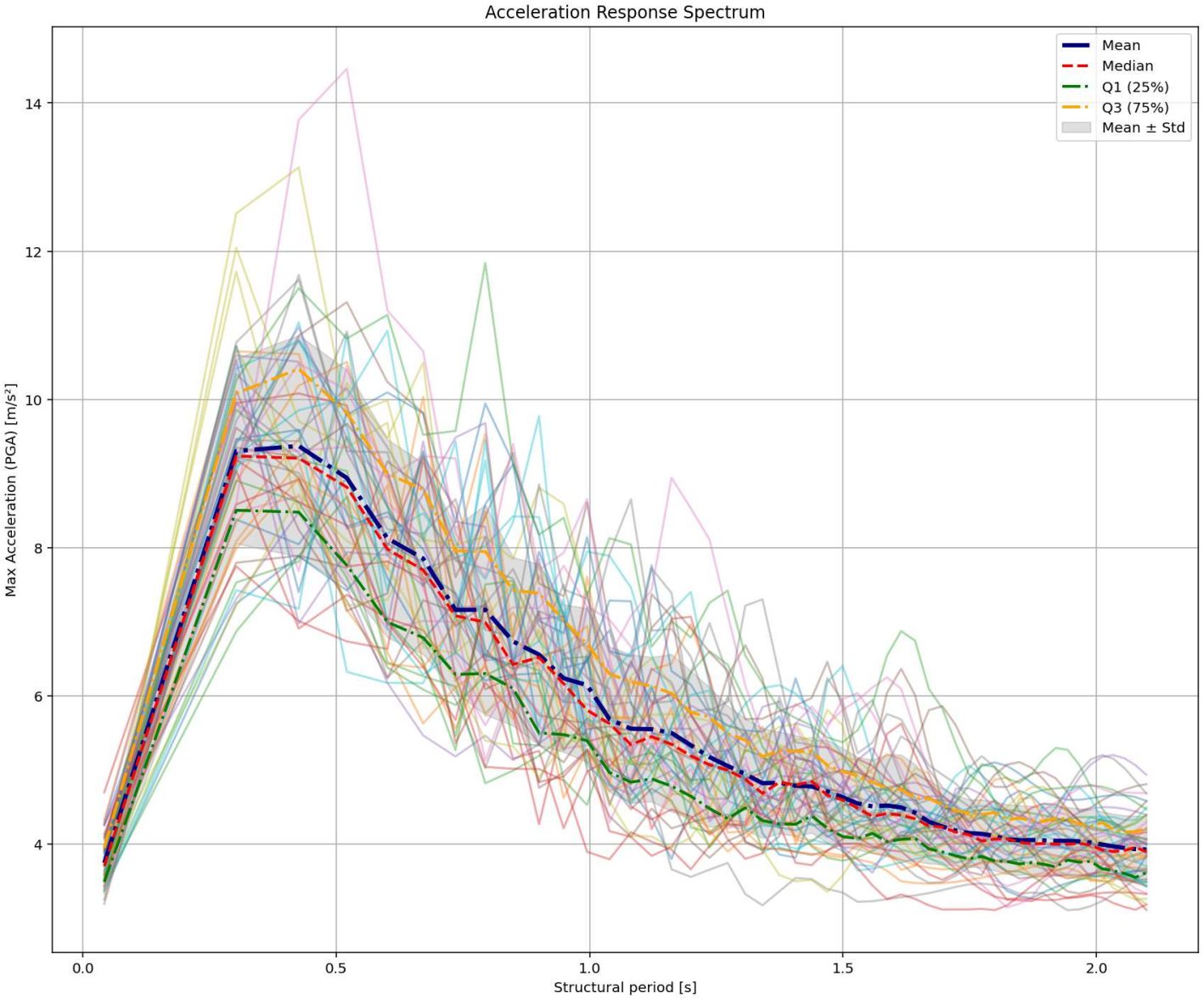

Cyclic Loading – Dissipated Energy
Base Shear (N)
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
$(u_{max}, F_{max})$
$E_d$ = 302221.470 N·m

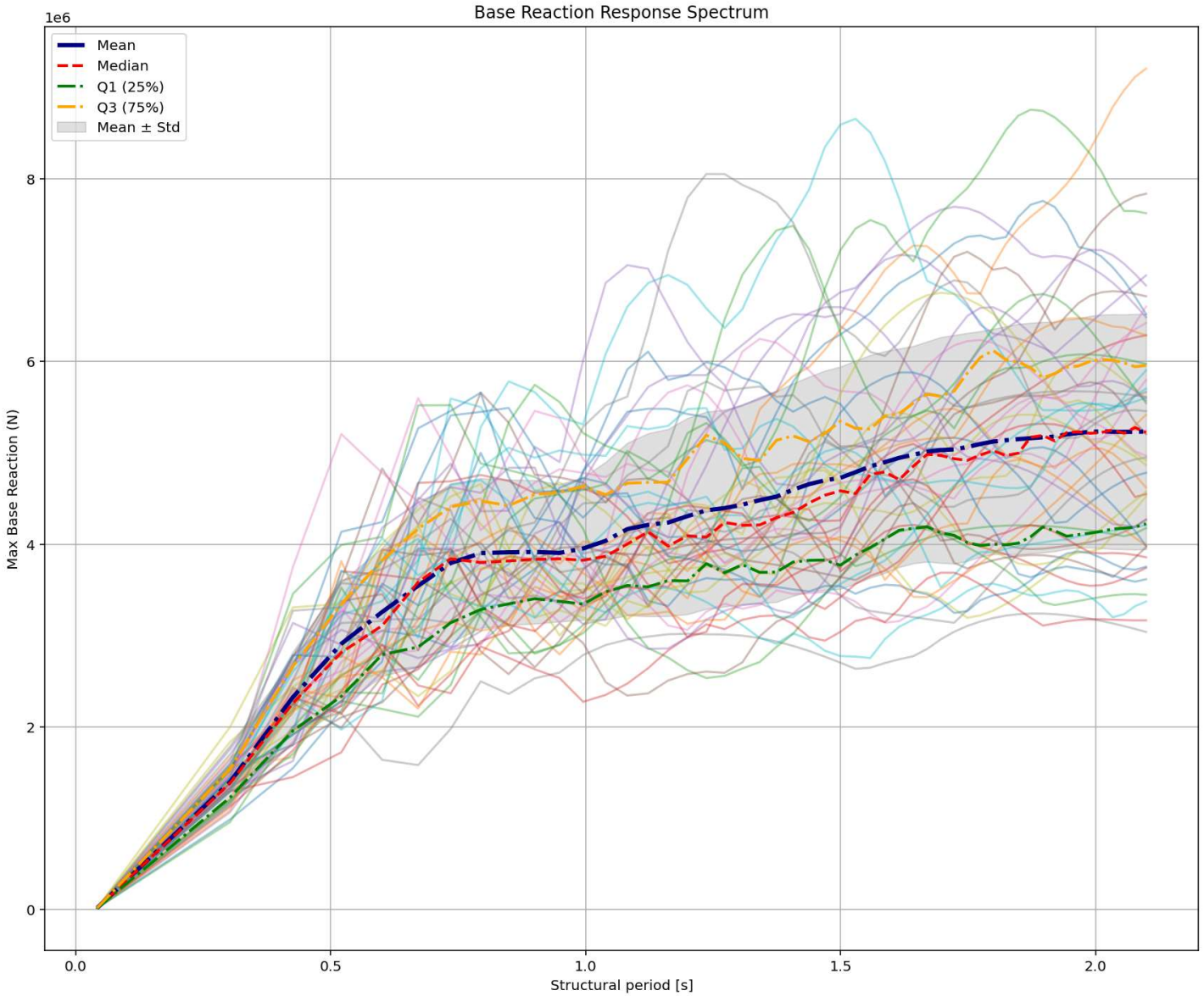

Earthquake Response – Dissipated Energy
1e6
6
4
2
0
−2
−4
−6
Base Shear (N)
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Displacement (m)
Hysteresis Loop
$(u_{max}, F_{max})$
$E_d$ = 701396.817 N·m

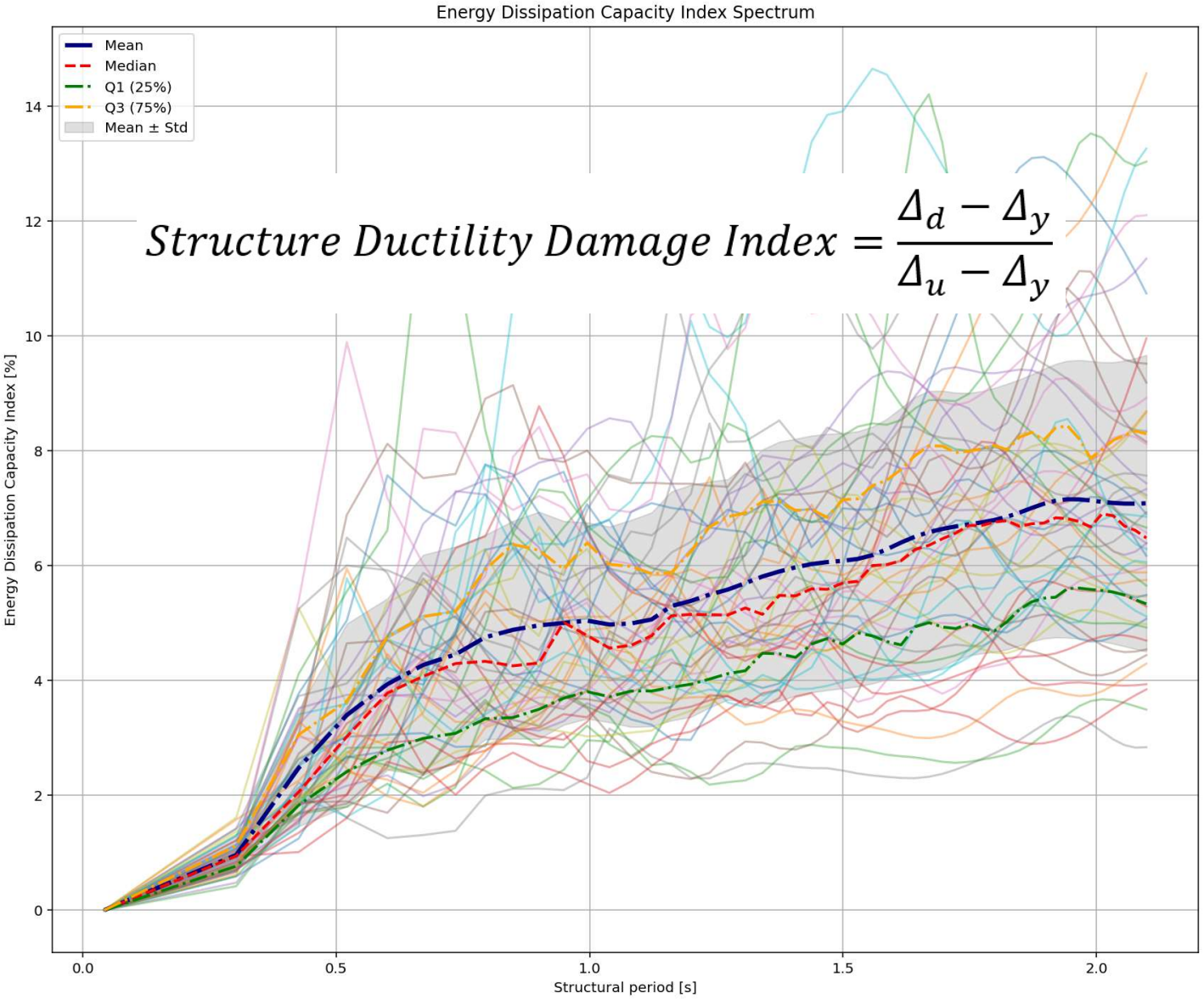

Equivalent viscous damping ratio - Incremental Dynamic Analysis Hysteresis
1e6
$\xi_{eq} = 17.4842$ [%]
$E_d = 498395.782$ N·m
$E_s = 226839.380$ N·m
Base Reaction [N]
Displacement [m]
Hysteresis Loop
$(u_{max}, F_{max})$
$E_d$ = 498395.782 N·m

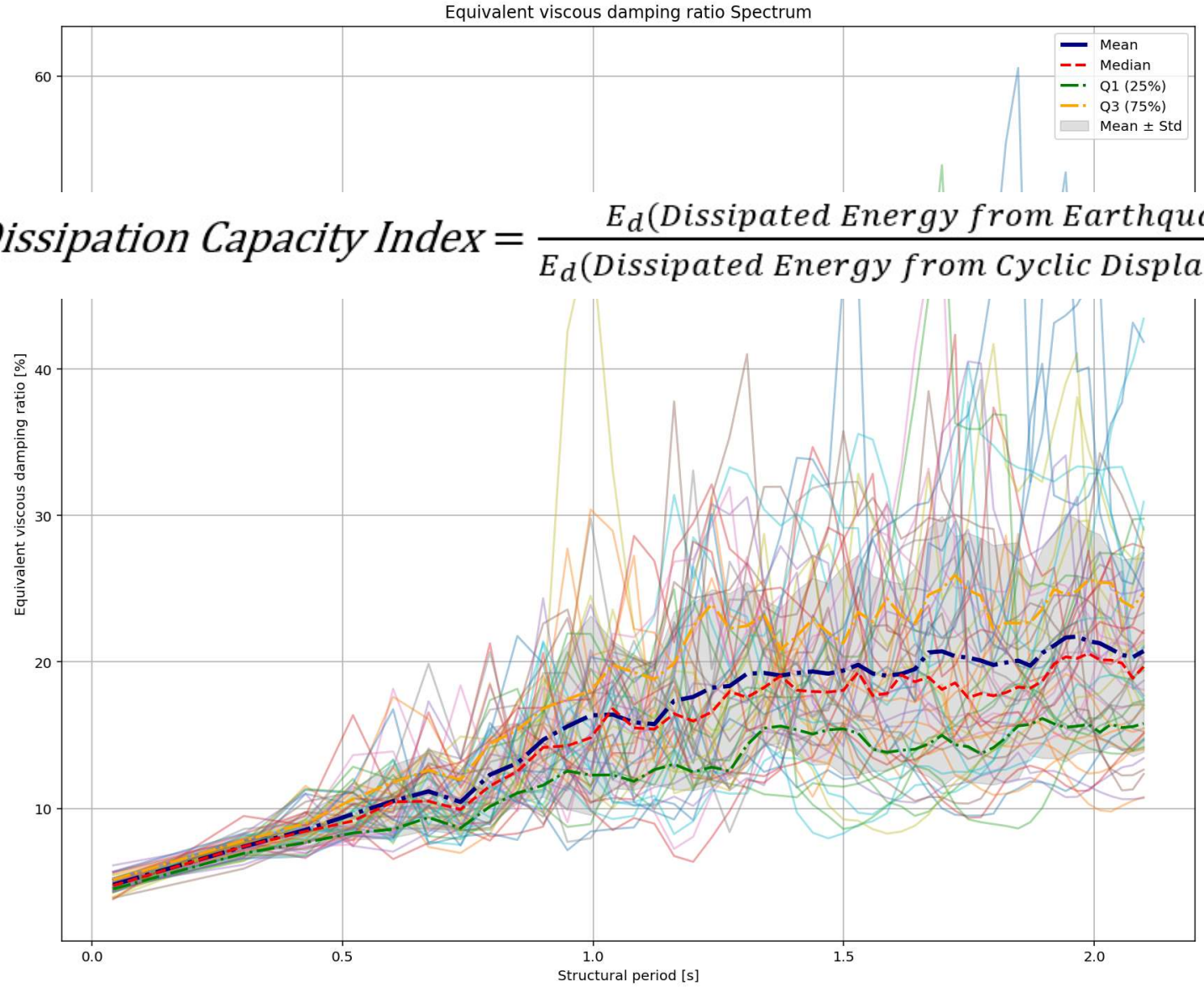

Displacement Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Displacement (PGD) [m]
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.0
0.5
1.0
1.5
2.0
Structural period [s]


Velocity Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Velocity (PGV) [m/s]
Structural period [s]
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0

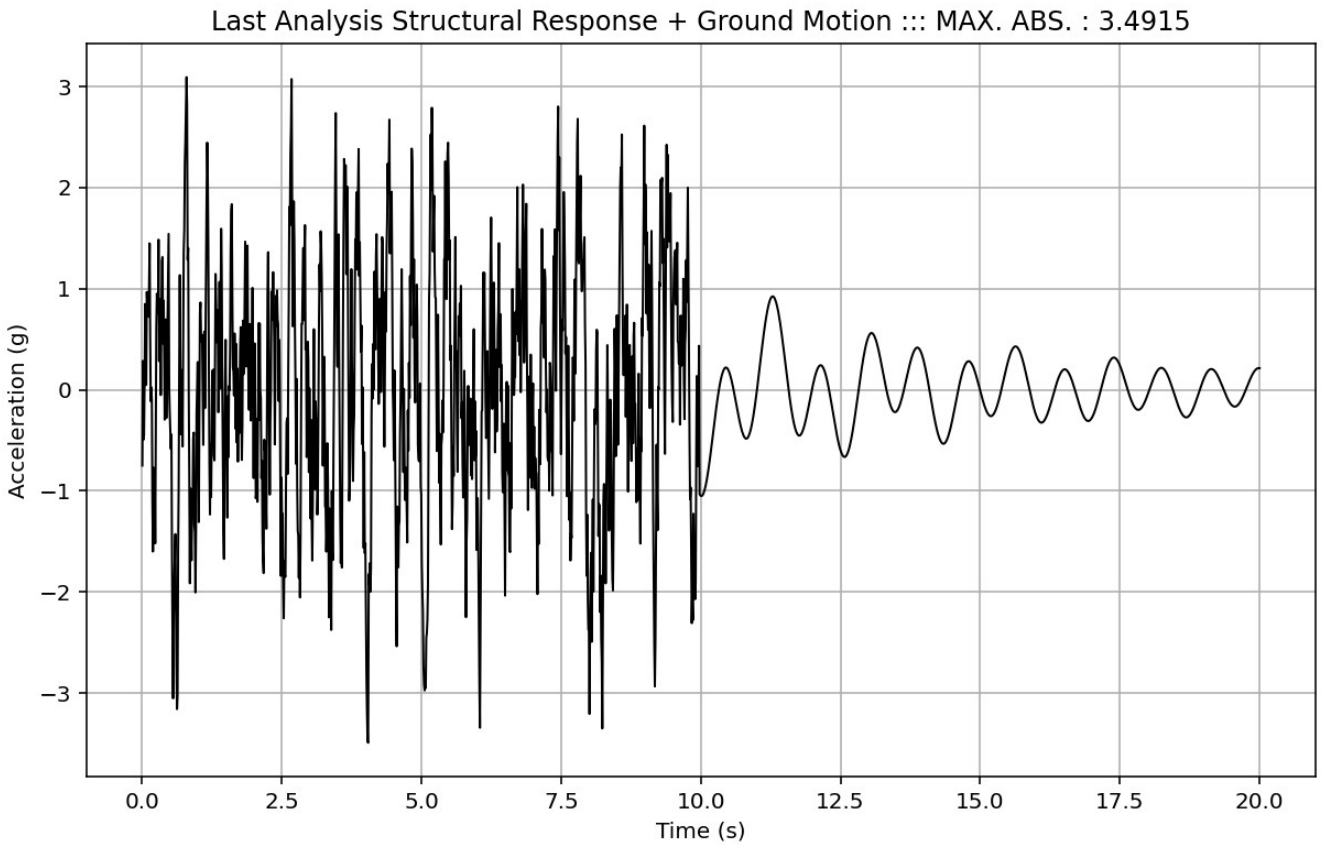

Acceleration Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Acceleration (PGA) [m/s²]
Structural period [s]
14
12
10
8
6
4
0.0
0.5
1.0
1.5
2.0

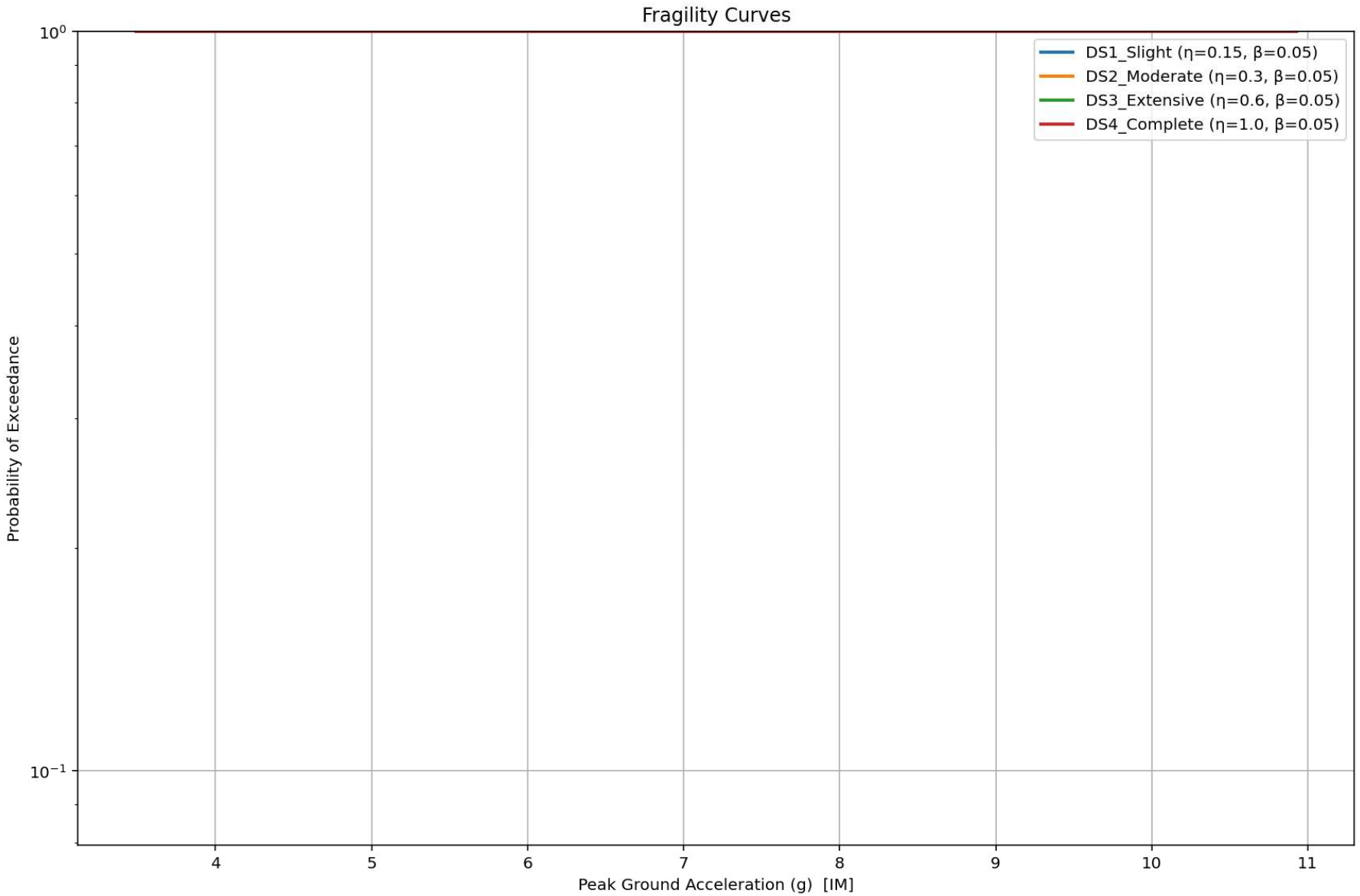

Base Reaction Response Spectrum
1e6
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Base Reaction (N)
Structural period [s]
0
2
4
6
8
0.0
0.5
1.0
1.5
2.0

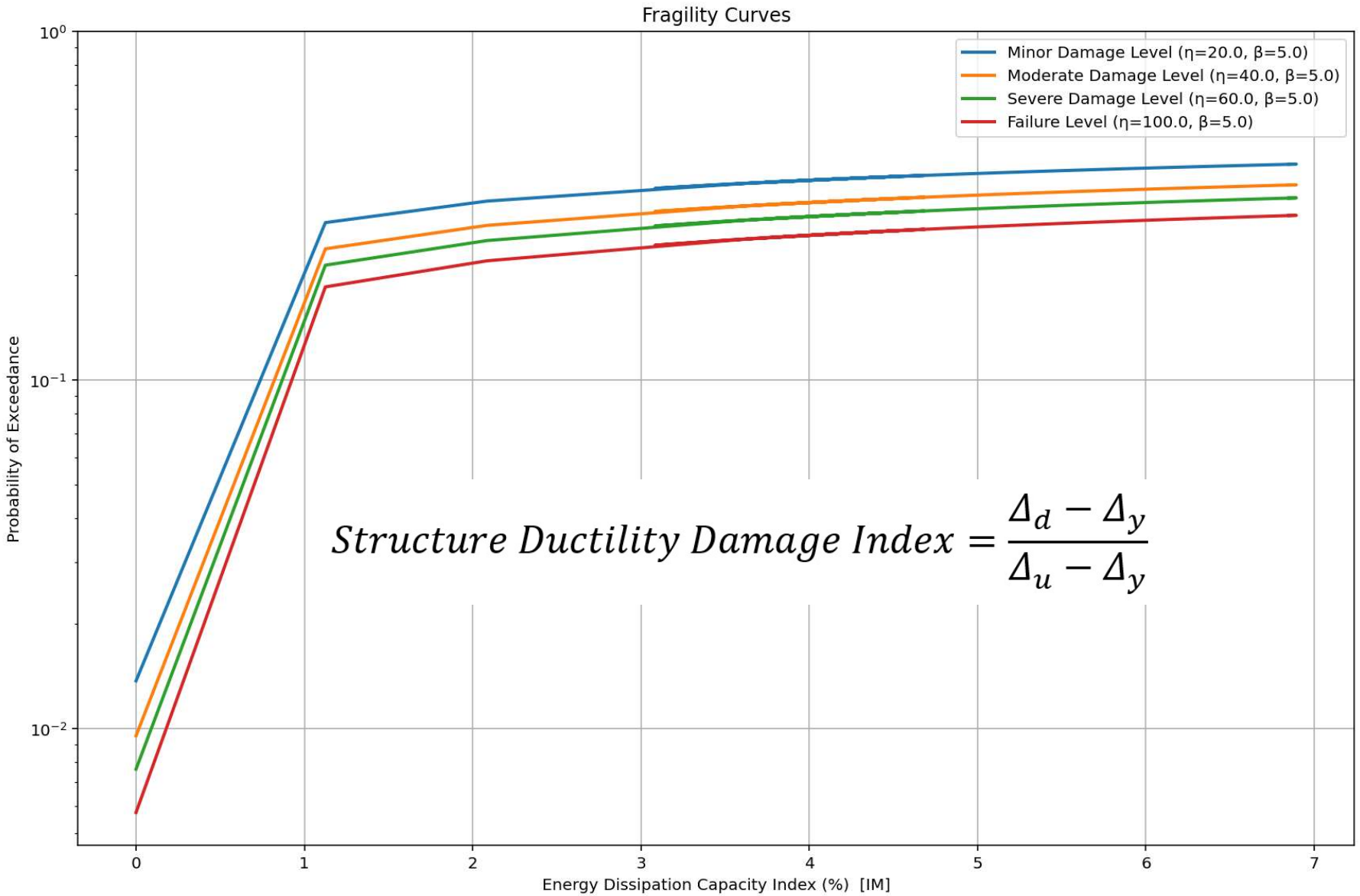

Energy Dissipation Capacity Index Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Structure Ductility Damage Index = $\frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$
Energy Dissipation Capacity Index [%]
Structural period [s]
0
2
4
6
8
10
12
14
0.0
0.5
1.0
1.5
2.0

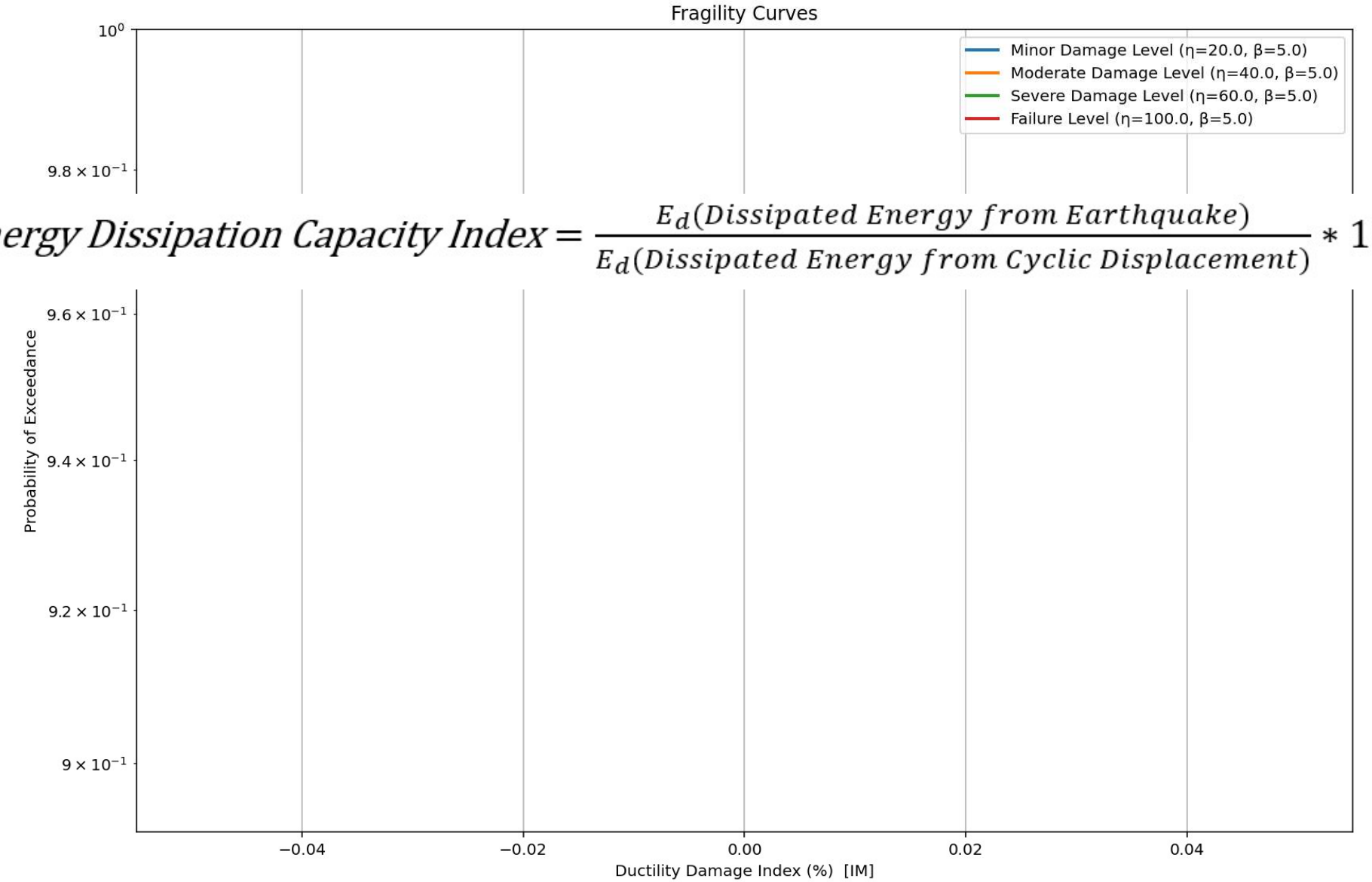



$$Energy\ Dissipation\ Capacity\ Index = \frac{E_d(Dissipated\ Energy\ from\ Earthquake)}{E_d(Dissipated\ Energy\ from\ Cyclic\ Displacement)} * 100$$

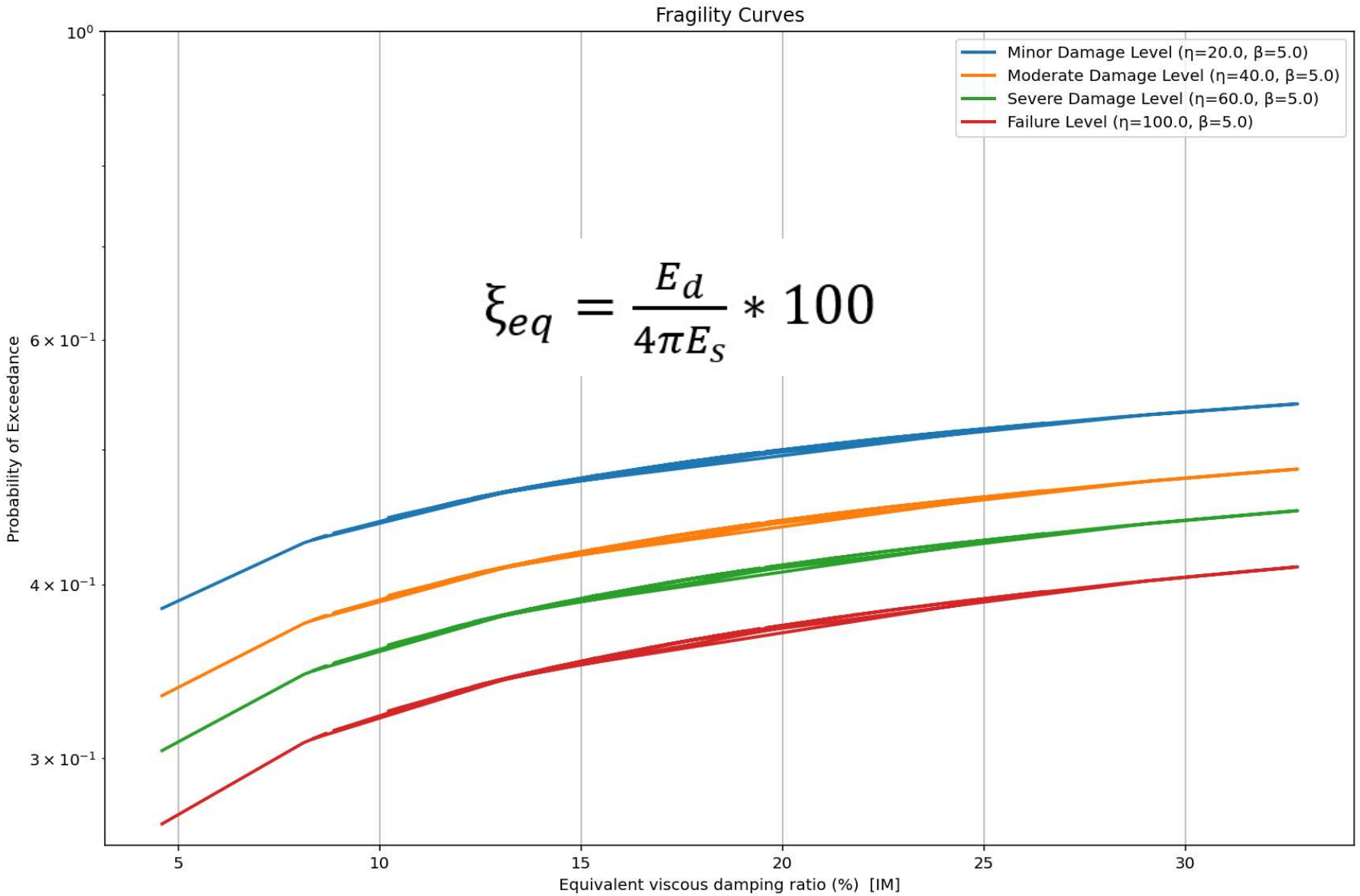

Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 3.4915
Acceleration (g)
Time (s)
3
2
1
0
−1
−2
−3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

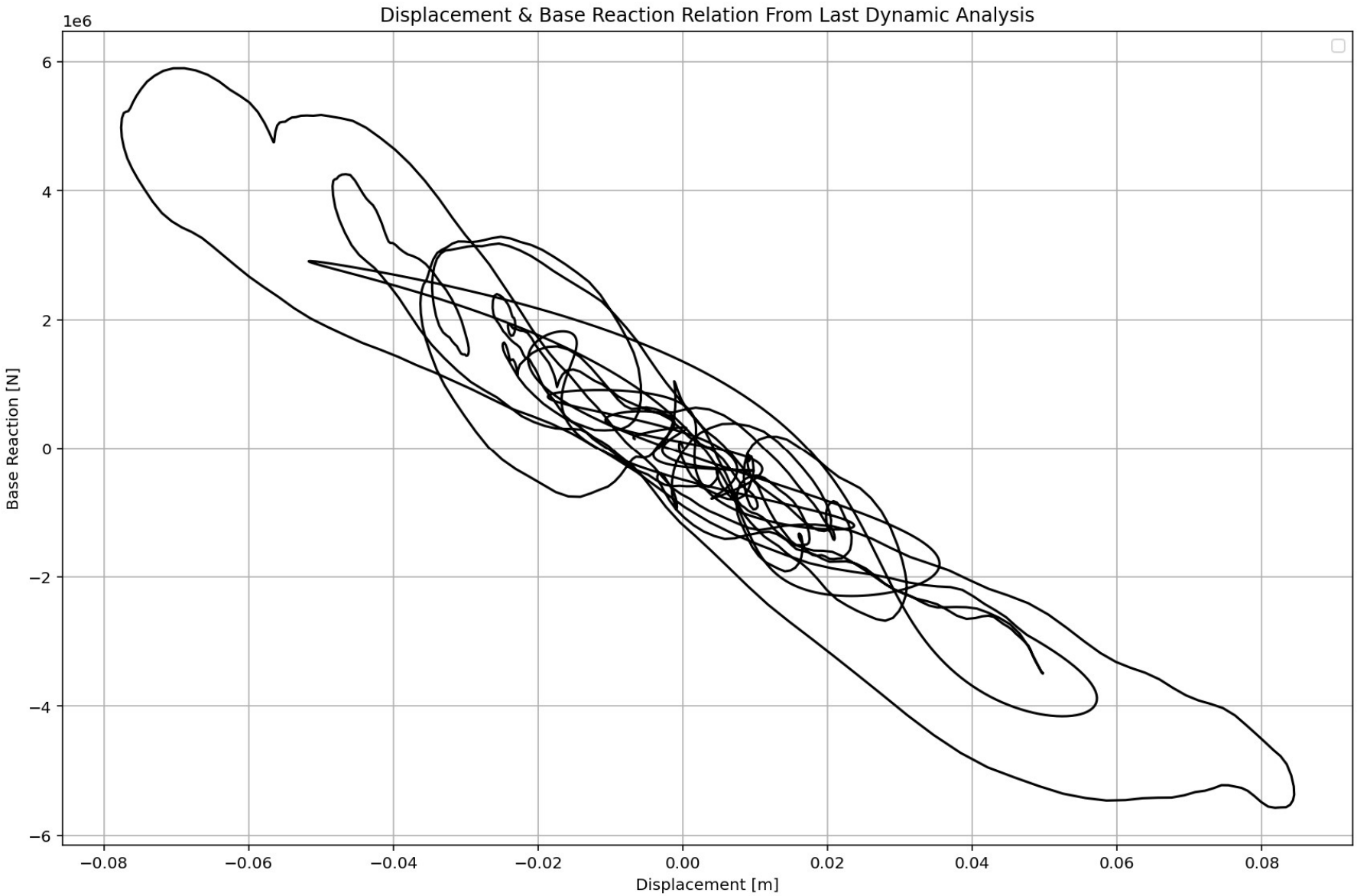

Fragility Curves
DS1_Slight (η=0.15, β=0.05)
DS2_Moderate (η=0.3, β=0.05)
DS3_Extensive (η=0.6, β=0.05)
DS4_Complete (η=1.0, β=0.05)
Probability of Exceedance
$10^{0}$
$10^{-1}$
4
5
6
7
8
9
10
11
Peak Ground Acceleration (g) [IM]

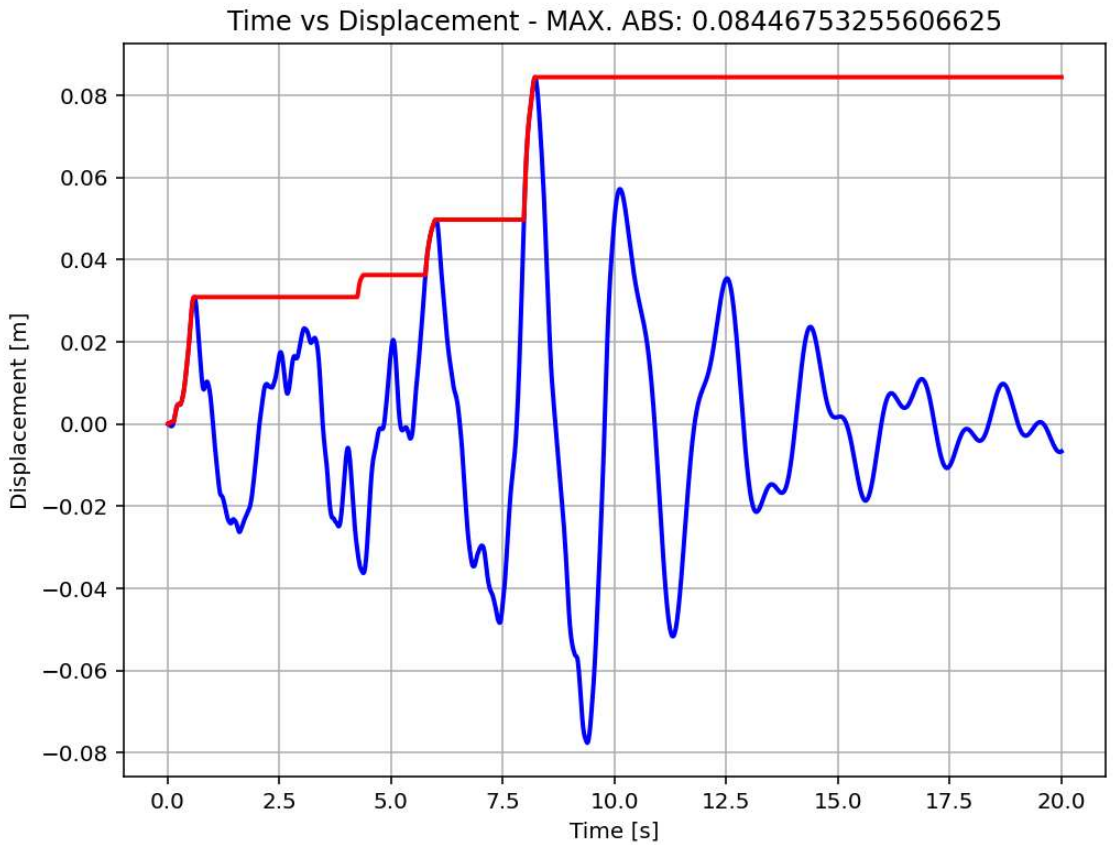

Fragility Curves
Minor Damage Level (η=20.0, β=5.0)
Moderate Damage Level (η=40.0, β=5.0)
Severe Damage Level (η=60.0, β=5.0)
Failure Level (η=100.0, β=5.0)
Probability of Exceedance
$10^0$
$10^{-1}$
$10^{-2}$
0
1
2
3
4
5
6
7
Energy Dissipation Capacity Index (%) [IM]
$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$

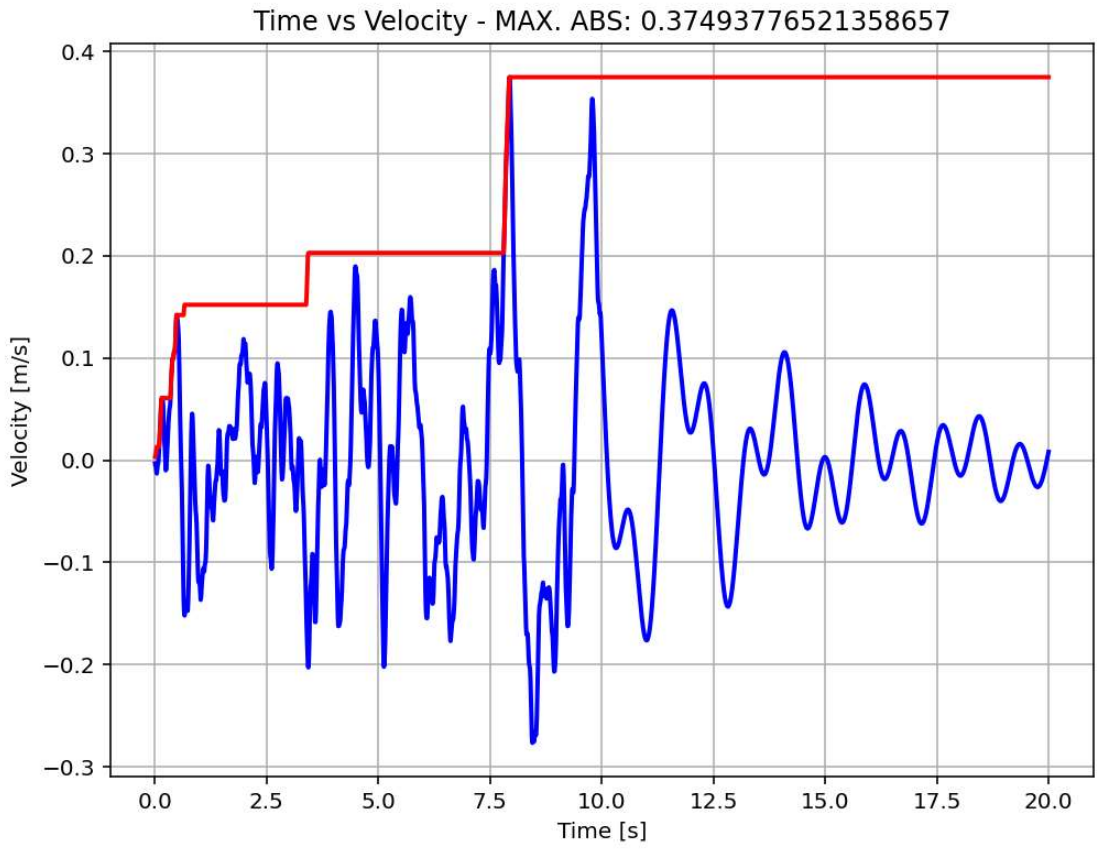


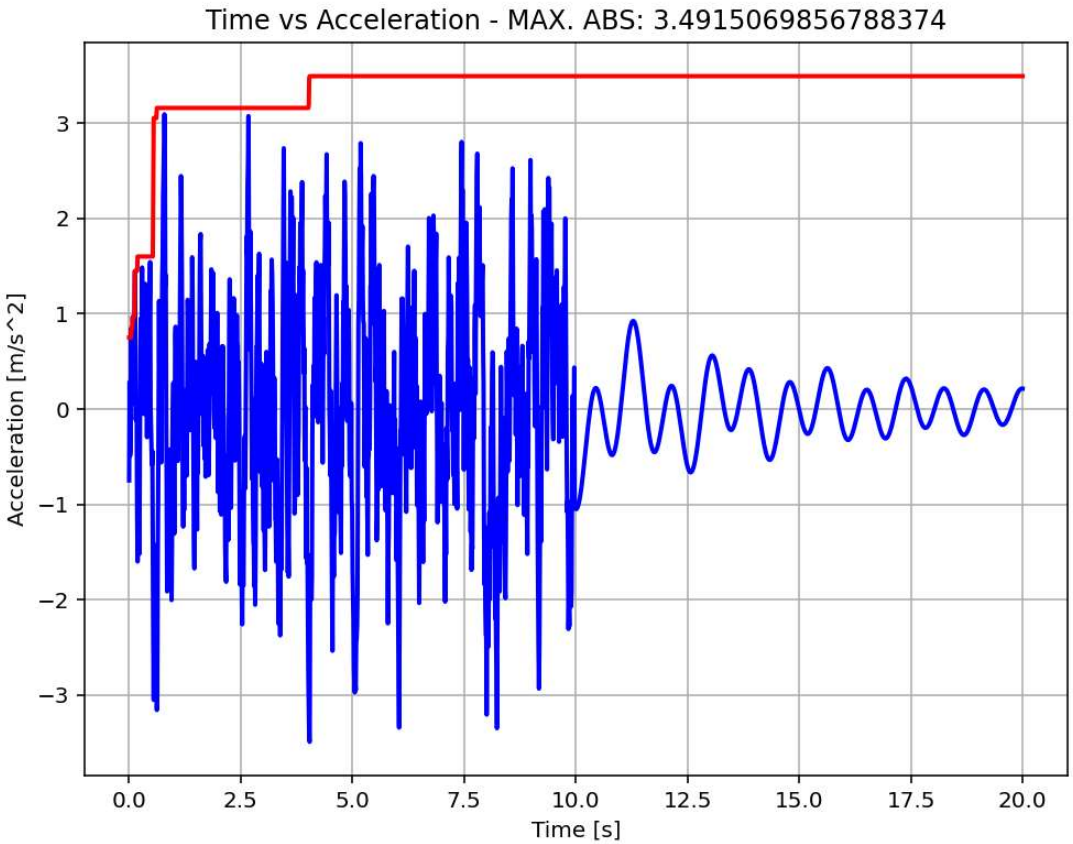

$$Energy\ Dissipation\ Capacity\ Index = \frac{E_d(Dissipated\ Energy\ from\ Earthquake)}{E_d(Dissipated\ Energy\ from\ Cyclic\ Displacement)} * 100$$

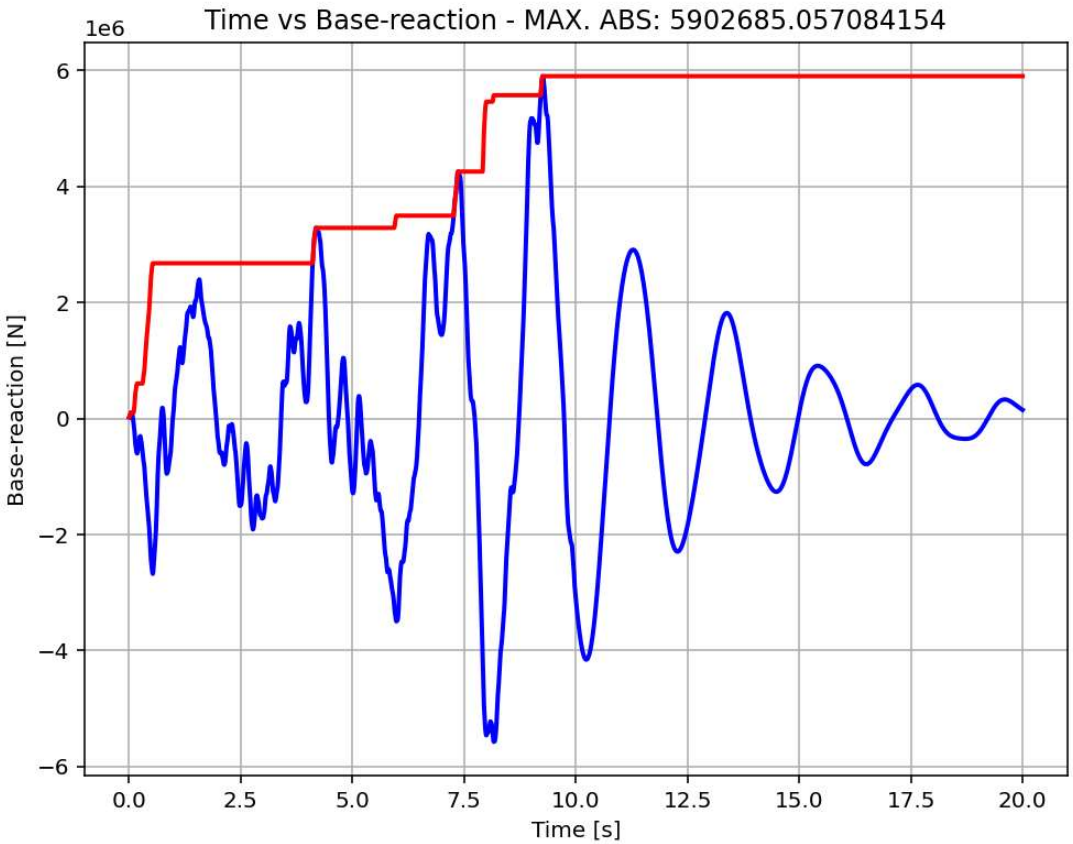

Fragility Curves
Minor Damage Level (η=20.0, β=5.0)
Moderate Damage Level (η=40.0, β=5.0)
Severe Damage Level (η=60.0, β=5.0)
Failure Level (η=100.0, β=5.0)
$\xi_{eq} = \frac{E_d}{4\pi E_s} * 100$
Probability of Exceedance
$10^0$
$6 \times 10^{-1}$
$4 \times 10^{-1}$
$3 \times 10^{-1}$
5
10
15
20
25
30
Equivalent viscous damping ratio (%) [IM]

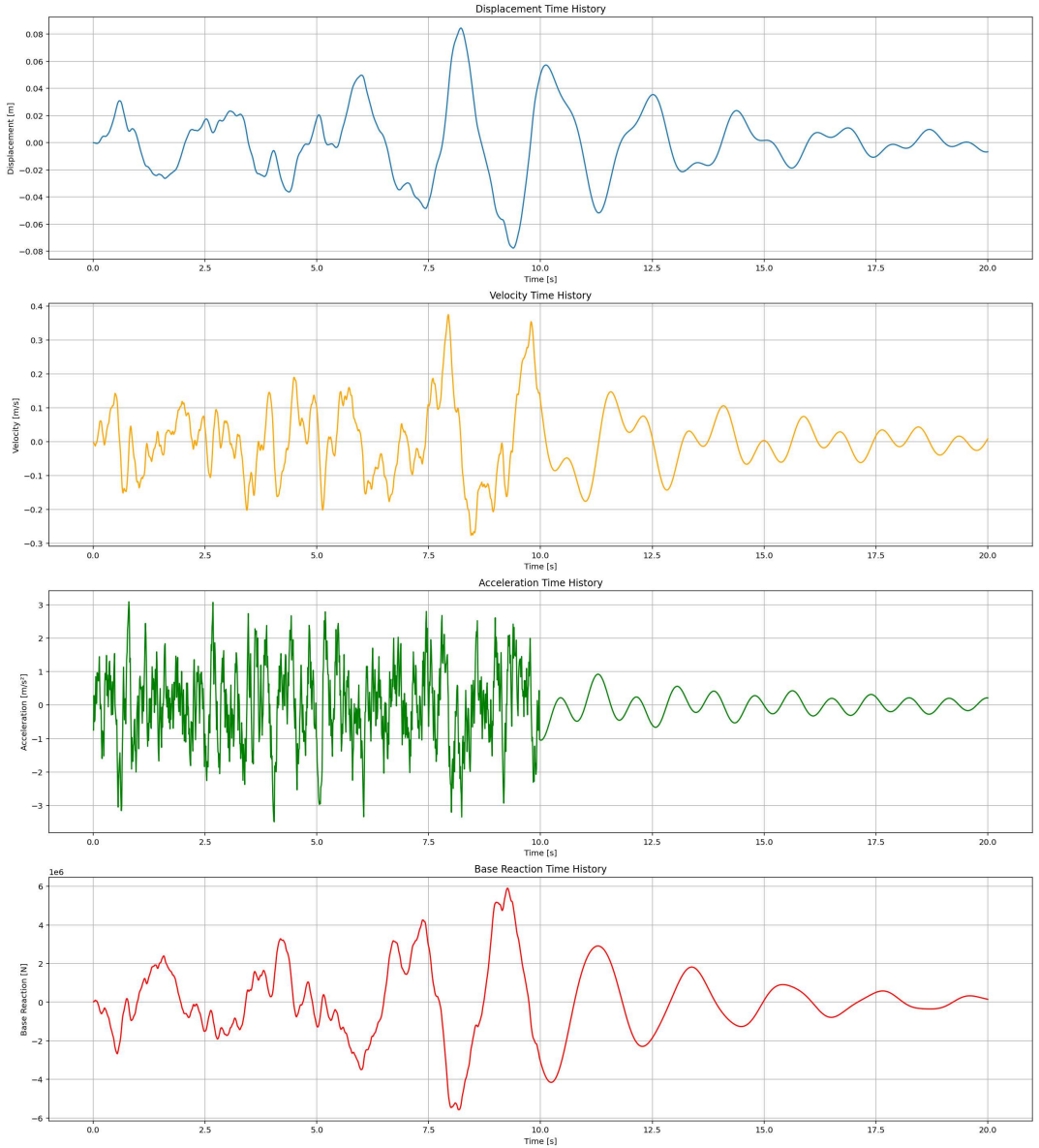

Displacement & Base Reaction Relation From Last Dynamic Analysis
1e6
6
4
2
0
−2
−4
−6
Base Reaction [N]
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Displacement [m]

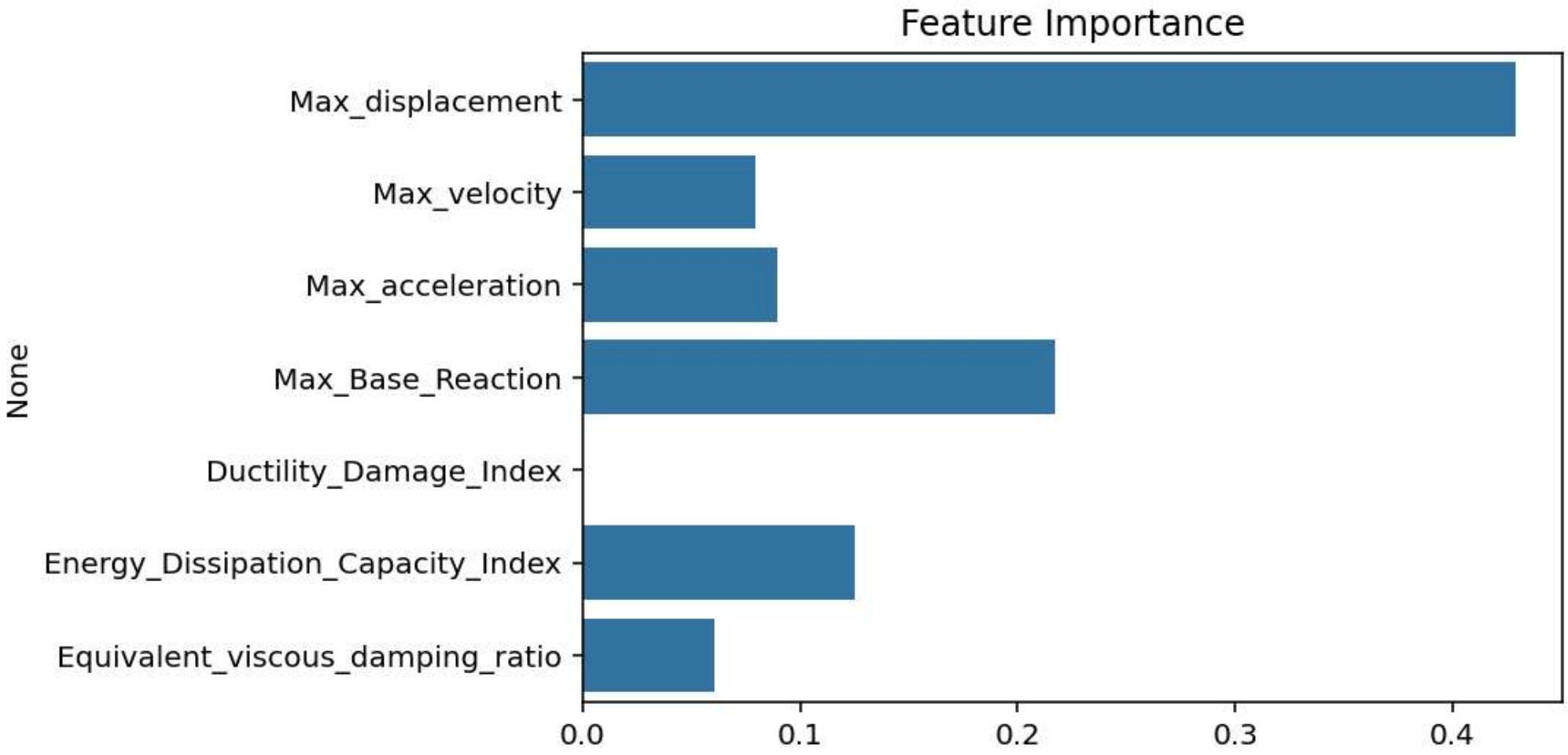

Time vs Displacement - MAX. ABS: 0.08446753255606625
Displacement [m]
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

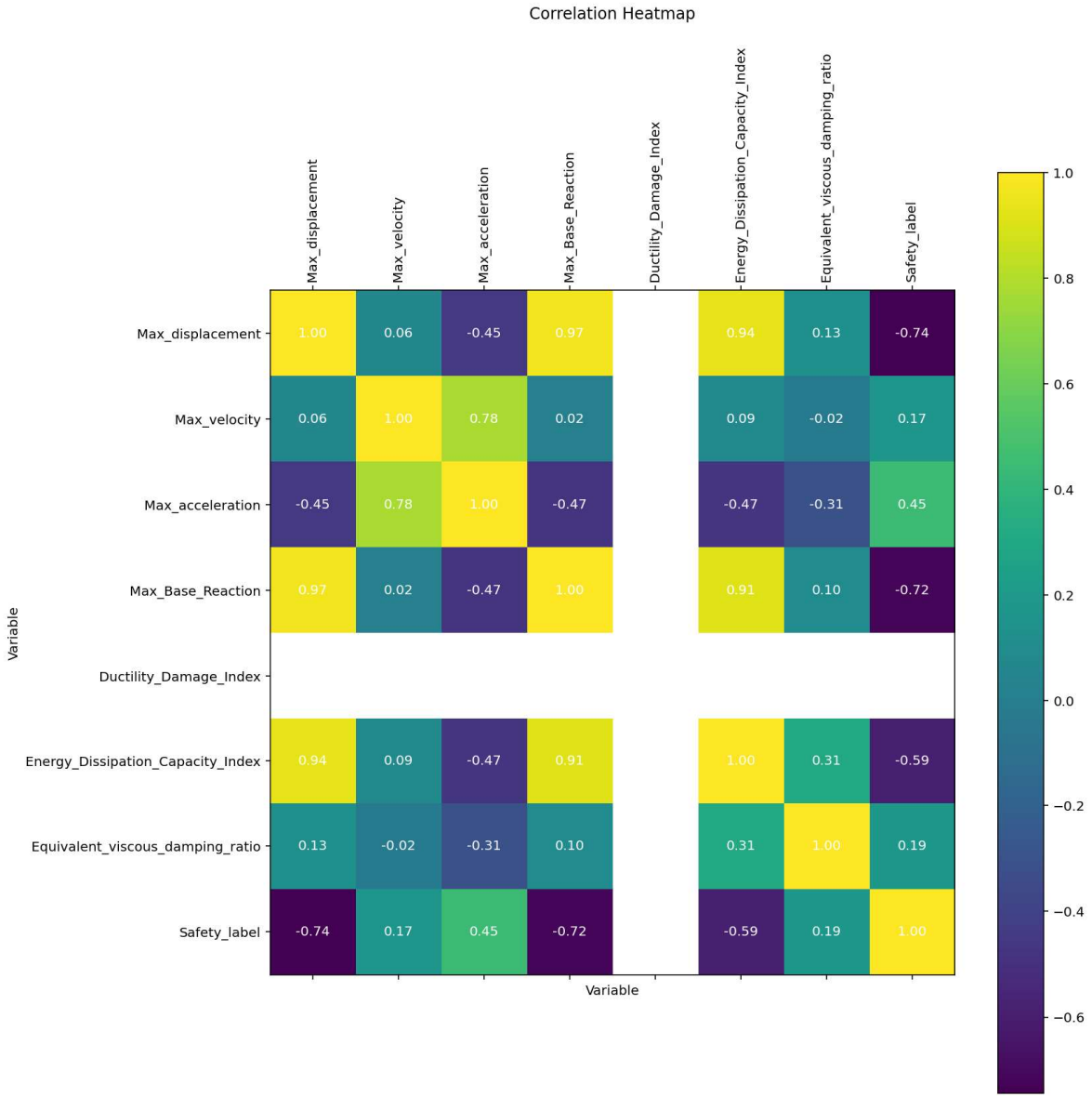

Time vs Velocity - MAX. ABS: 0.37493776521358657
Velocity [m/s]
Time [s]
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Time vs Acceleration - MAX. ABS: 3.4915069856788374
Acceleration [m/s^2]
Time [s]
3
2
1
0
−1
−2
−3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Time vs Base-reaction - MAX. ABS: 5902685.057084154
1e6
6
4
2
0
−2
−4
−6
Base-reaction [N]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Displacement Time History
Displacement [m]
Time [s]
Velocity Time History
Velocity [m/s]
Time [s]
Acceleration Time History
Acceleration [m/s²]
Time [s]
Base Reaction Time History
1e6
Base Reaction [N]
Time [s]


Feature Importance
Max_displacement
Max_velocity
Max_acceleration
Max_Base_Reaction
Ductility_Damage_Index
Energy_Dissipation_Capacity_Index
Equivalent_viscous_damping_ratio
None
0.0
0.1
0.2
0.3
0.4


Correlation Heatmap
Max_displacement
Max_velocity
Max_acceleration
Max_Base_Reaction
Ductility_Damage_Index
Energy_Dissipation_Capacity_Index
Equivalent_viscous_damping_ratio
Safety_label
Variable
Variable
1.00 0.06 -0.45 0.97 0.94 0.13 -0.74
0.06 1.00 0.78 0.02 0.09 -0.02 0.17
-0.45 0.78 1.00 -0.47 -0.47 -0.31 0.45
0.97 0.02 -0.47 1.00 0.91 0.10 -0.72
0.94 0.09 -0.47 0.91 1.00 0.31 -0.59
0.13 -0.02 -0.31 0.10 0.31 1.00 0.19
-0.74 0.17 0.45 -0.72 -0.59 0.19 1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6